CalCOFI

The purpose of this research paper is to provide a comprehensive, step-by-step guide for conducting a **Repeated Measures ANOVA** using the Isalos Analytics Platform. Repeated Measures ANOVA extends one-way ANOVA by analyzing **multiple measurements** taken from the same subjects (or casts) across different conditions or time points, allowing analysts to assess how within-subject factors (e.g., depth) and between-subject factors (e.g., cast) influence a set of interrelated continuous outcomes. This methodology is particularly valuable in oceanographic research, where environmental parameters such as temperature and salinity are measured at various depths and should be analyzed together to account for within-cast correlations.

Repeated Measures ANOVA combines the principles of multivariate ANOVA (MANOVA) and univariate repeated measures. By including within-subject factors (e.g., Depth), the procedure accounts for the correlation between repeated measurements, thereby increasing statistical power and providing more accurate estimates of differences across conditions while controlling for individual variability.

Isalos version used: 2.1.5

The dataset used in this analysis is the **CalCOFI (California Cooperative Oceanic Fisheries Investigations) reduced Dataset**, available on Kaggle.

Source: [CalCOFI]

This dataset simulates real-world oceanographic records and includes cast-level information such as bottle count, cast count, depth measurements, temperature, and salinity. It is designed to support environmental monitoring and ecological research.

For the purposes of this Repeated Measures ANOVA analysis, variables are classified as follows:

- **Dependent Variables (Responses):**

  - *T_degC* — Temperature in degrees Celsius or
  - *Salnty* — Salinity in practical salinity units

  These are continuous outcome measures to be analyzed simultaneously across depths.

- **Within-Subject Factor:**
  - *Depthm* — Depth in meters (e.g., 0, 10, 20, 30, 50, 75 m)

  This categorical variable represents the repeated-measures condition across which each cast is measured.

- **Between-Subject Factor:**
  - *Cst_Cnt* — Cast count (e.g., 1 through 6)

  This categorical variable represents independent groups whose multivariate effect on the oceanographic outcomes is of primary interest.

Step 1: Import Data from file

Firstly, launch the Isalos Analytics Platform (version 2.1.2). Then, in the left spreadsheet panel, create a new tab by clicking **File** and choose **Open** to load the CalCOFI dataset (or simply copy-paste the data from your spreadsheet).

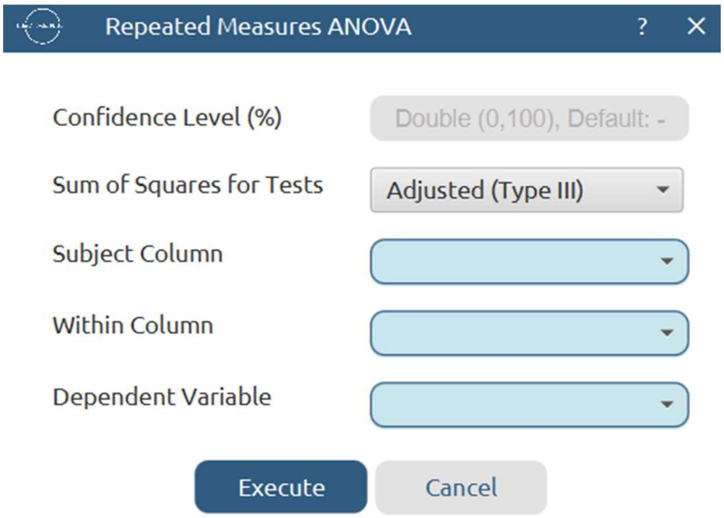
Btl_Cnt	Cst_Cnt	Depthm	T_degC	Salnty
1	1	0	10.5	33.44
2	1	10	10.46	33.437
3	1	20	10.45	33.421
4	1	30	10.45	33.431
5	1	50	10.24	33.424
6	1	75	9.86	33.494
7	1	100	9.67	33.58
8	2	0	10.1	32.95
9	2	10	9.89	32.94
10	2	20	9.75	32.939
11	2	30	9.52	33.013
12	2	50	9.04	33.272
13	2	75	8.83	33.488
14	2	100	8.26	33.631
15	3	0	10.2	32.63
16	3	10	10.05	32.66

Step 2: Access the RM-ANOVA function

Navigate to the analysis module through the main menu by choosing the following path:

Statistics → Analysis of (Co)Variance → RM ANOVA

Then, you will see the RM ANOVA configuration window:



Step 3: Configure RM ANOVA parameters

The Repeated Measures ANOVA model can be specified with varying levels of complexity. The model is expressed as:

$$[T\_degC, Salnty] = \mu + \alpha_i(Cst\_Cnt) + \beta_j(Depthm) + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

This is a multivariate repeated measures model where the two dependent variables are regressed onto the within-subject factor (Depth) and the between-subject factor (Cast). The model includes main effects and the interaction between Cast and Depth.

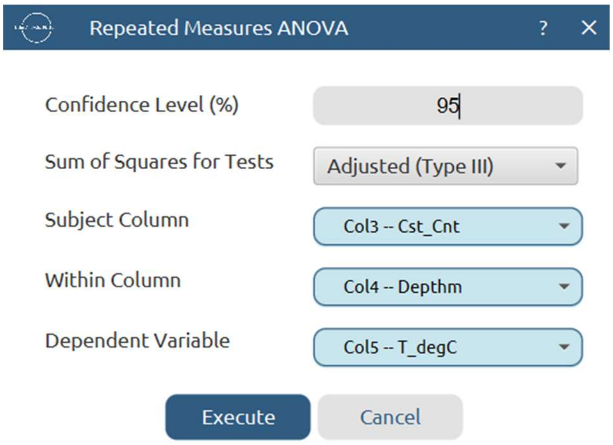
Important: The inclusion of interaction terms (e.g., Cst_Cnt × Depthm) should be guided by theoretical considerations and exploratory data analysis. Repeated Measures ANOVA assumes sphericity—if this assumption is violated, corrections (e.g., Greenhouse-Geisser) may be necessary. For this basic demonstration, we focus on the **full factorial model**.

Steps in Isalos Analytics Platform:

In the RM ANOVA configuration window, set the following parameters:

* **Confidence Level (%):** 95
* **Subject Column:** Col3 -- Cst_Cnt
* **Within Column:** Col4 -- Depthm
* **Dependent Variable**: Col5 -- T_degC

The formula preview will display the multivariate model as shown below:



After configuring all parameters:

* Review all selections to ensure accuracy.
* Click the **Execute** button to run the analysis.
* Wait for the computation to complete (processing time depends on dataset size and the number of dependent variables).

Step 4: Results

The analysis results will be displayed in the right spreadsheet panel. The output typically includes the following components:

WITHIN-SUBJECTS EFFECTS							
Source	Depthm	DF	Adj SS	Adj MS	F-Value	P-Value	Significant
Depthm	Linear	1	6.5331175	6.5331175	21.3491685	0.0012534	Yes
Error(Depthm)	Linear	9	2.7541146	0.3060127			
Depthm	Quadratic	1	1.1805001	1.1805001	12.8985658	0.0058255	Yes
Error(Depthm)	Quadratic	9	0.8236963	0.0915218			
Depthm	Cubic	1	0.0601667	0.0601667	1.1751302	0.3065340	No
Error(Depthm)	Cubic	9	0.4608000	0.0512000			
Depthm	Quartic	1	0.0056894	0.0056894	0.2649781	0.6191230	No
Error(Depthm)	Quartic	9	0.1932392	0.0214710			
Depthm	Order 5	1	0.0001458	0.0001458	0.0345801	0.8566020	No
Error(Depthm)	Order 5	9	0.0379554	0.0042173			
Depthm	Order 6	1	0.0043091	0.0043091	2.7022689	0.1346180	No
Error(Depthm)	Order 6	9	0.0143516	0.0015946			
BETWEEN-SUBJECTS EFFECTS							
Source	DF	Adj SS	Adj MS	F-Value	P-Value	Significant	
Intercept	1	8260.1165714	8260.1165714	892.4651812	0E-7	Yes	
Error	9	83.2985429	9.2553937				

Step 5 : Residual Analysis & Assumption's checks

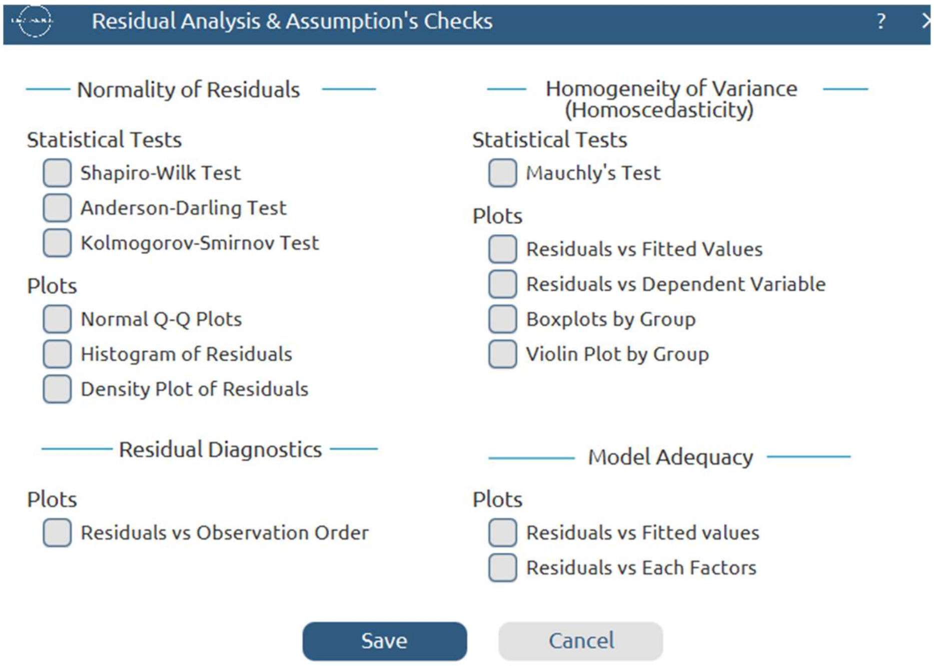
Choose

Statistics → Analysis of (Co)Variance → RM ANOVA

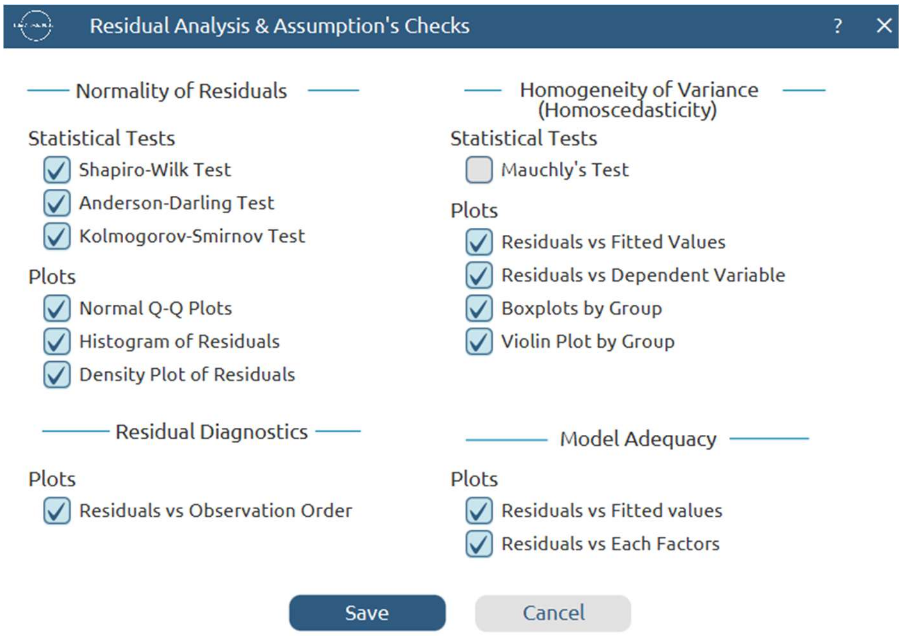
and choose

Residual Analysis & Assumption's Checks.

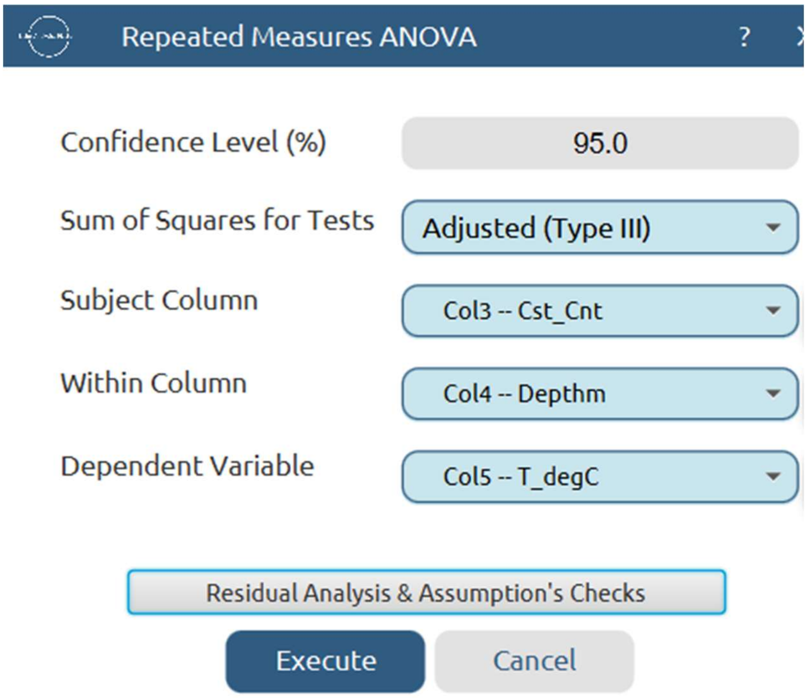
You get the following window:



and then choose the tests you want. For example, choose the following:



and click save. You should have the following window open:



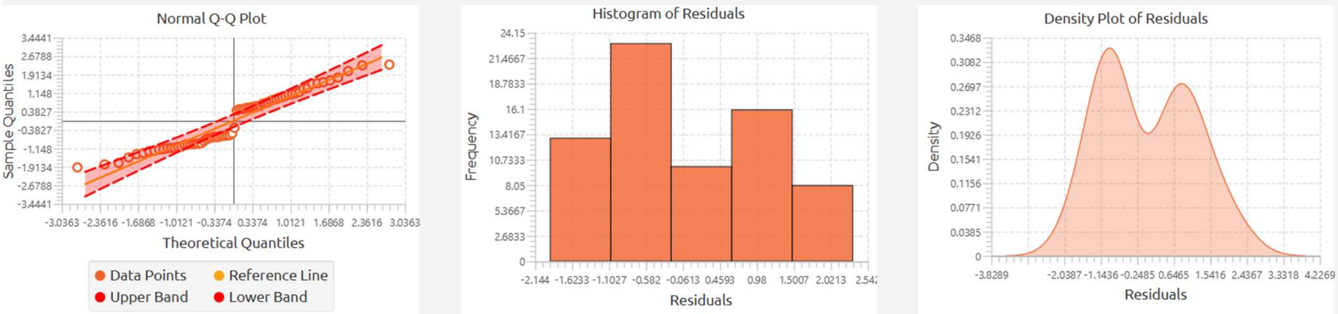
After that, click execute and you get the results

Repeated Measures ANOVA

Normality of Residuals

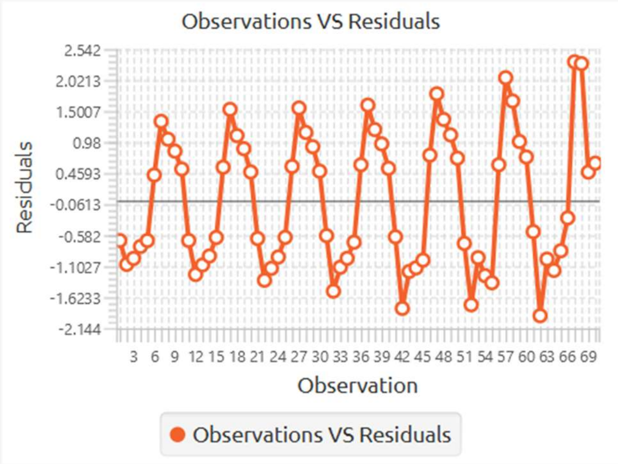
Statistical Tests

Normality Test	Statistic	Prob	P-value	Decision
Shapiro Wilk	0.934	0.999	0.001	Reject Null Hypothesis
Anderson Darling	2.053	1.000	0.000	Reject Null Hypothesis
Kolmogorov Smirnov	0.185	0.986	0.014	Reject Null Hypothesis

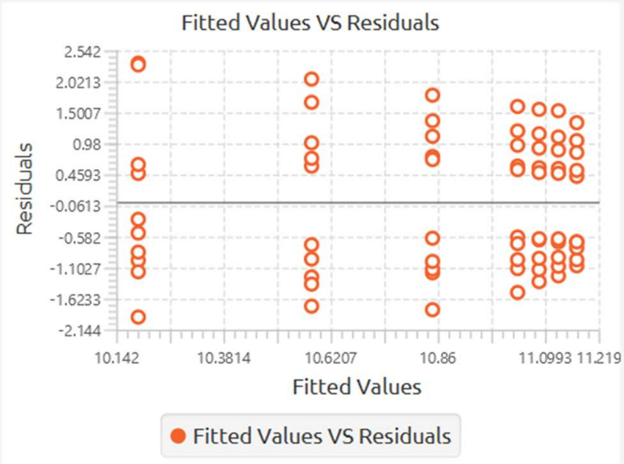


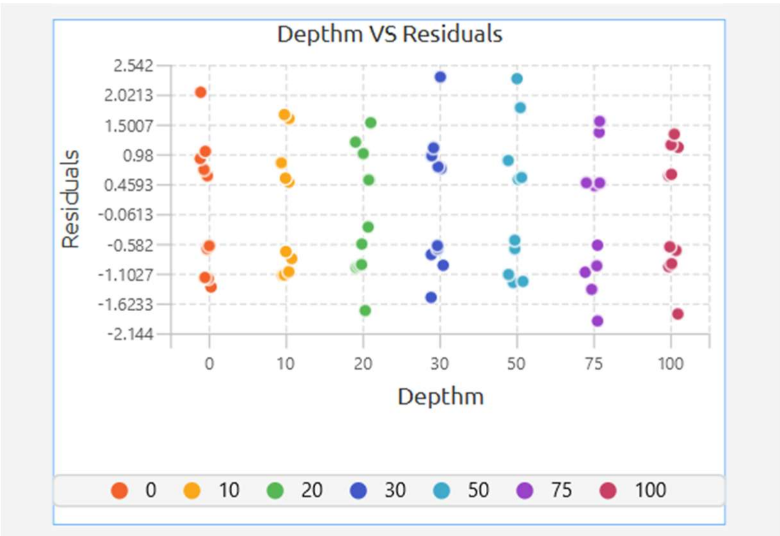

Homogeneity of Variance
Fitted Values VS Residuals
Residuals
Fitted Values
Fitted Values VS Residuals
T_degC VS Residuals
Residuals
T_degC
T_degC VS Residuals
Depthm VS T_degC
T_degC
Depthm
Depthm VS T_degC
T_degC
Depthm

Residual Diagnostics



Model Adequacy





Part 2: Descriptive Statistics

This step clarifies the central tendency, variability, and distribution of temperature and salinity across depths and casts. It also helps to detect any unusual values, check the balance of the design, and provide a clear context for the inferential tests that follow.

In this section, we present means, standard deviations, and minimum/maximum values for the two dependent variables – **T_degC** (temperature) and **Salnty** (salinity) – grouped by **Depthm** (within-subject factor) and by **Cst_Cnt** (between-subject factor). We also provide a cross-tabulation of means for each combination of depth and cast to reveal interaction patterns.

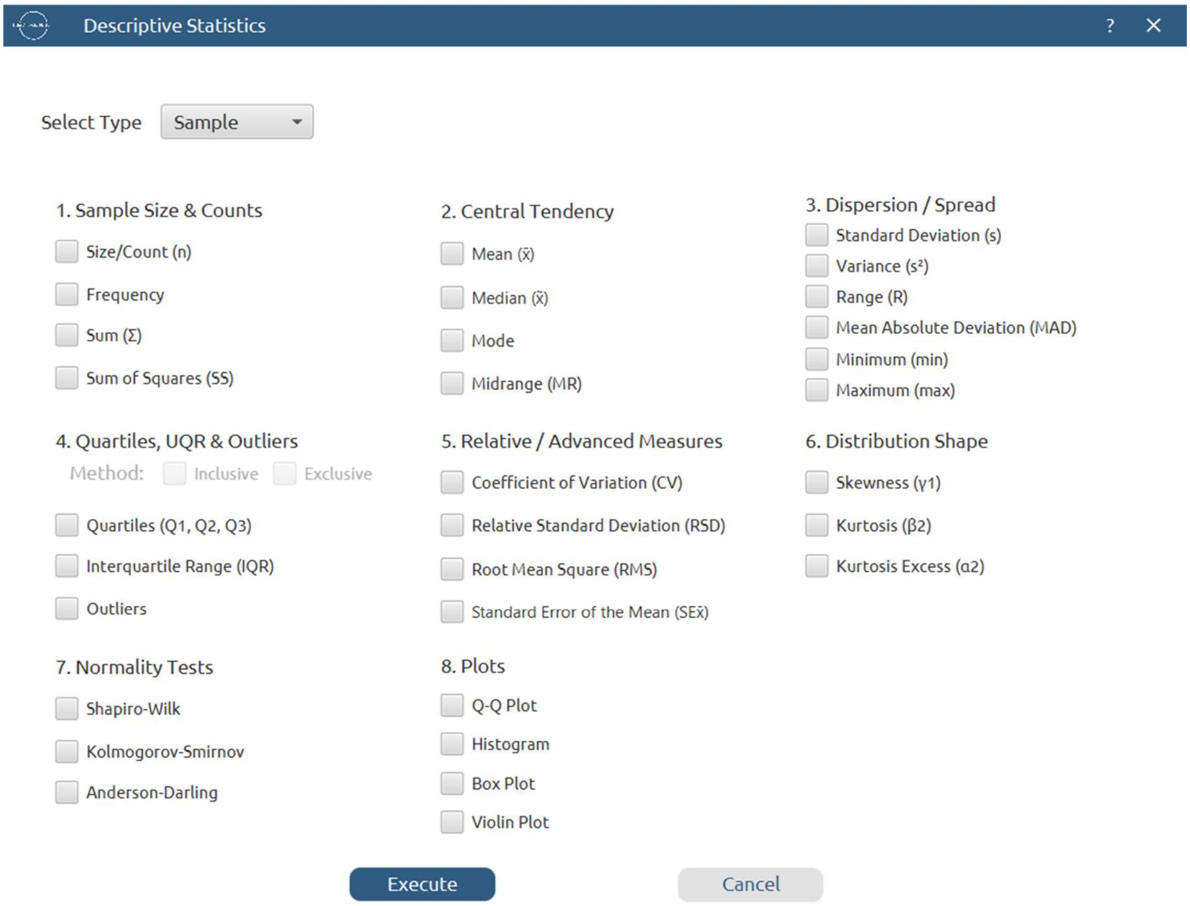
Step 1: Import the data (use the same data as before)

Step 2: Access the descriptive statistics function

Navigate to the descriptive analysis module through the main menu by choosing the following path:

Statistics → Basic Statistics → Descriptive Statistics

Then, you will see the descriptive statistics configuration window:



Step 3: Configure parameters

In the descriptive statistics window, set the following parameters:

1. Sample Size & Counts
    a. Size / Count (n)
    b. Frequency
    c. Sum
    d. Sum of Squares

2. Central Tendency
    a. Mean
    b. Median
    c. Mode
    d. Midrange

3. Dispersion / Spread
    a. Standard Deviation
    b. Variance
    c. Range
    d. Mean Absolute Deviation
    e. Minimum
    f. Maximum

4. Quartiles, UQE & Outliers
    a. Quartiles (Q1,Q2,Q3)
    b. Interquartile Range (IQR)
    c. Outliers

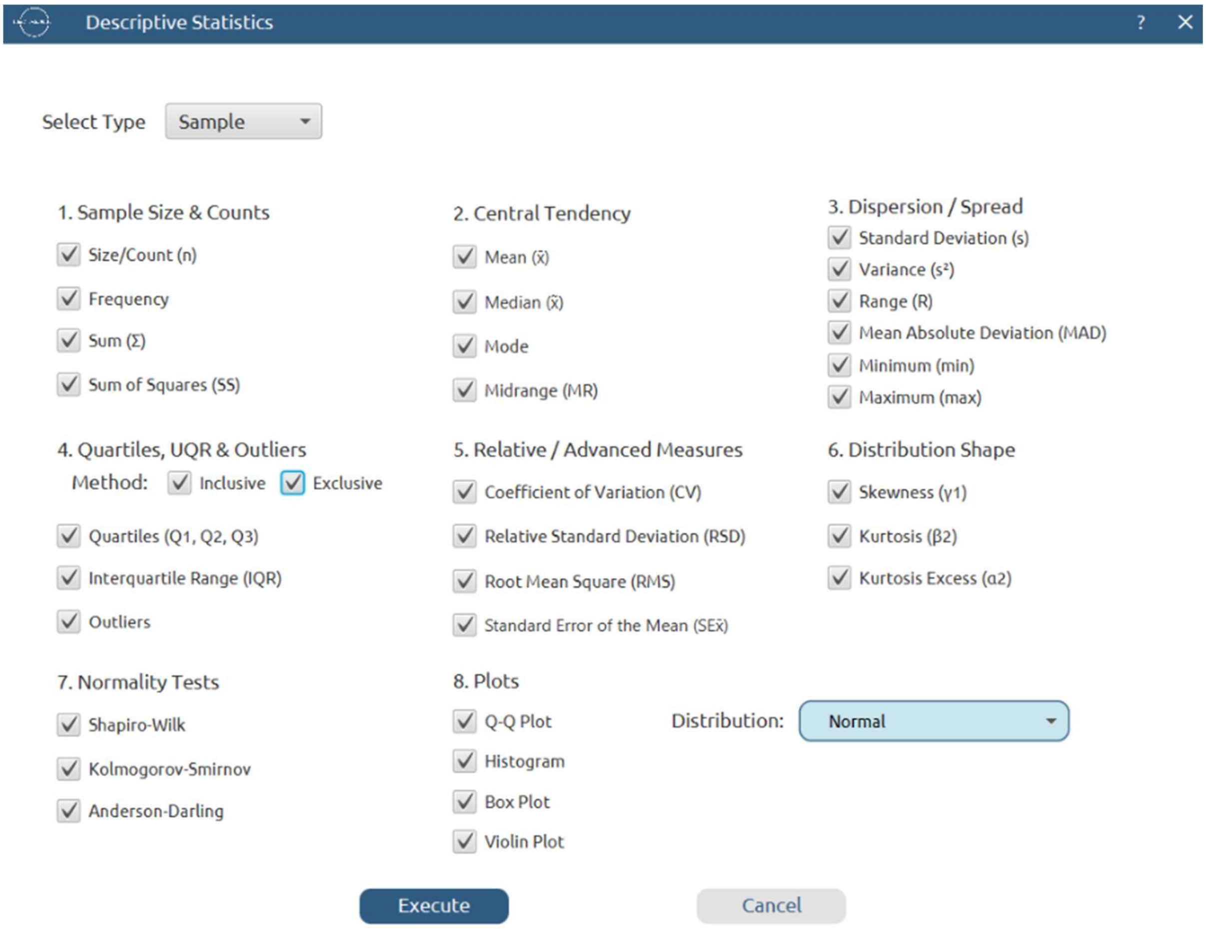
5. Relative / Advanced Measures
    a. Coefficient of Variation
    b. Relative Standard Deviation
    c. Root Mean Square
    d. Standard Error of the Mean

6. Distribution Shape
    a. Skewness
    b. Kurtosis
    c. Kurtosis excess

7. Normality Tests
    a. Shapiro-Wilk
    b. Kolmogorov-Smirnov
    c. Anderson-Darling

8. Plots
    a. Q-Q Plot
    b. Histogram
    c. Box Plot
    d. Violin Plot

The above preview will display the multivariate model as shown below:



After configuring all parameters:

- Review all selections to ensure accuracy.
- Click the **Execute** button to run the descriptive analysis.
- Wait for the computation to complete (processing time depends on dataset size and the number of dependent variables).

Step 4: Results

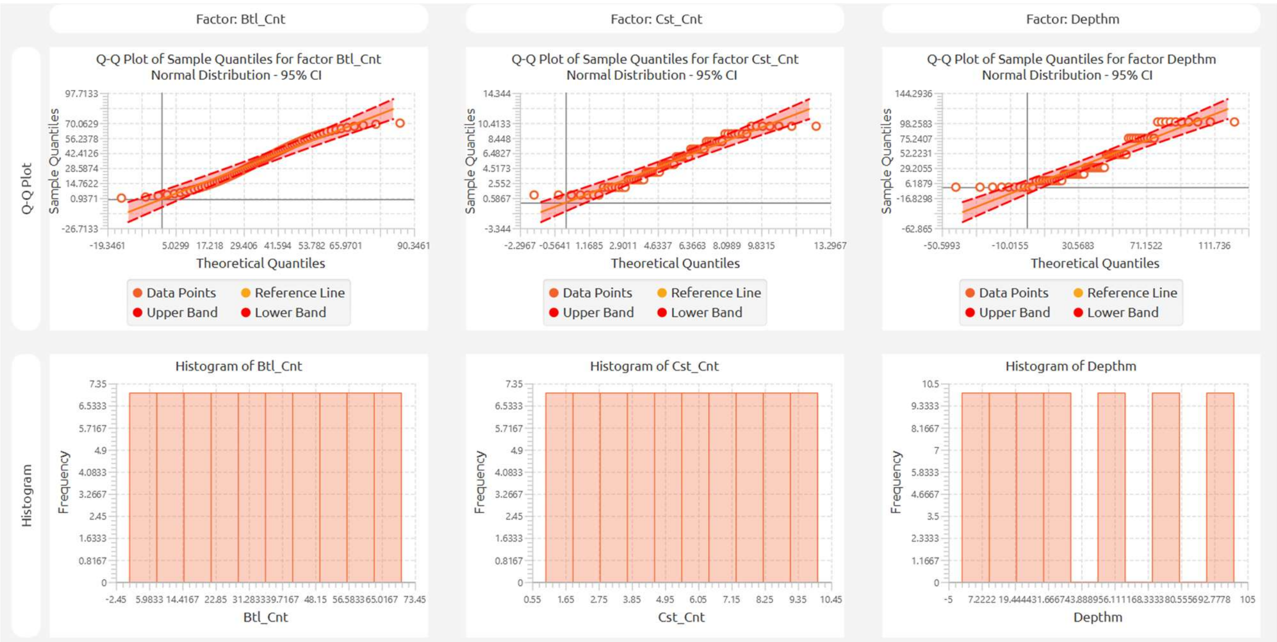
The analysis results will be displayed in the right spreadsheet panel. The output typically includes the following components:

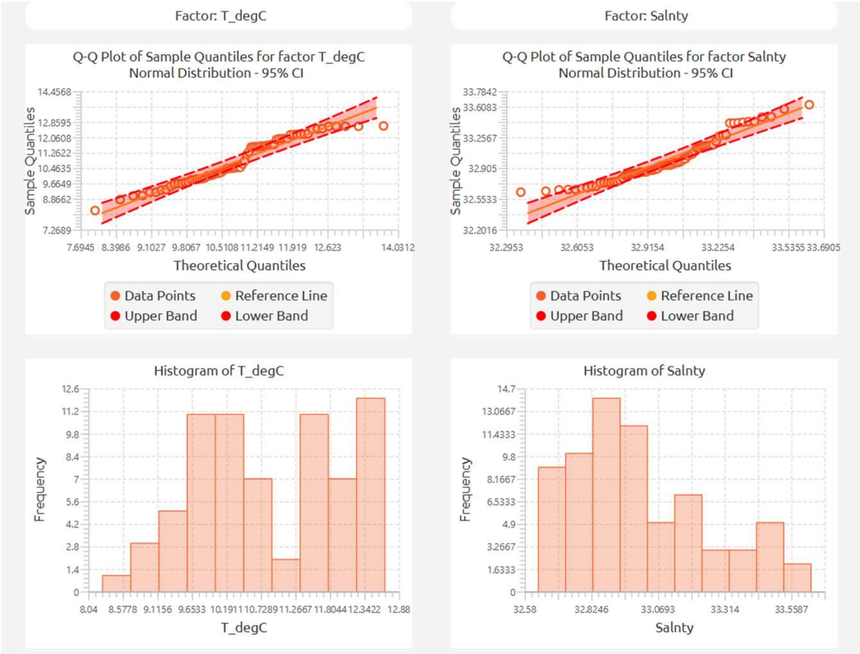
Factors	Size/Count	Sum	Sum of Squares	Mean	Median	Mode	Mid Range	Standard Deviation	Variance	Range
Btl_Cnt	70.0	2485.0	28577.5	35.5	35.5	None	35.5	20.3510851	414.1666667	69.0

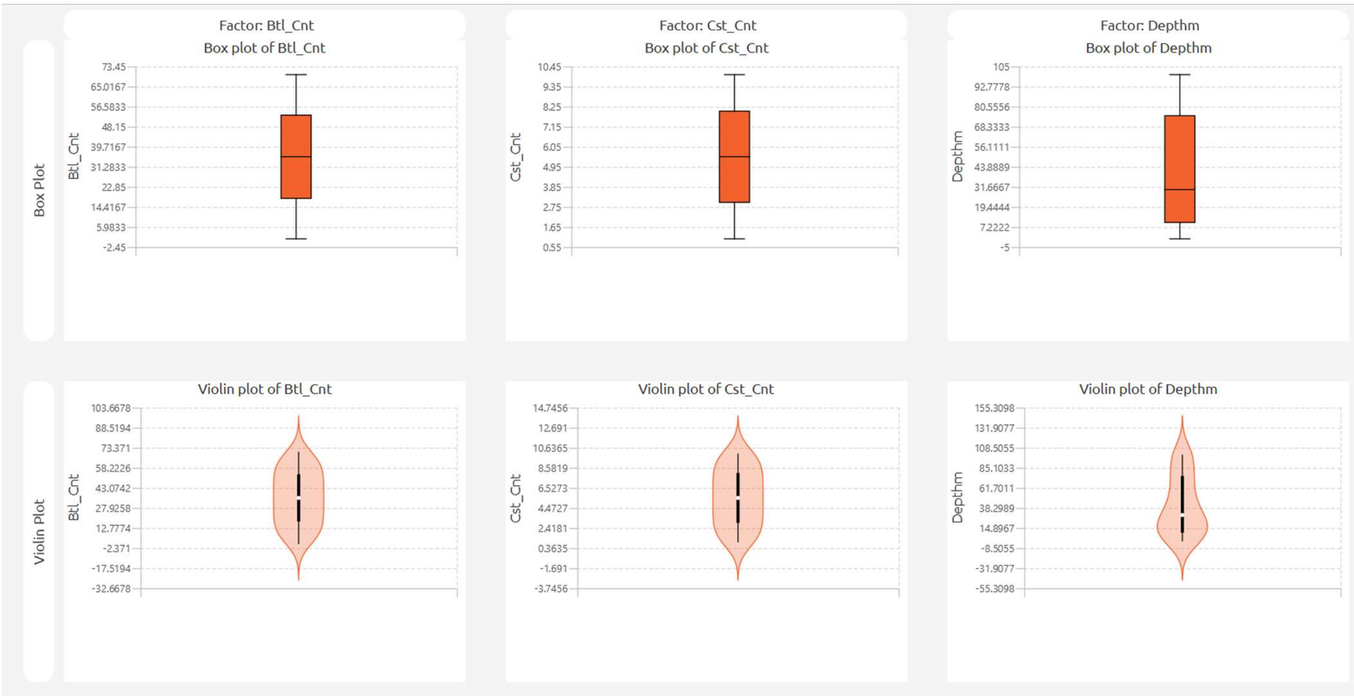
Mean Absolute Deviation	Minimum	Maximum	Quartiles - Inclusive	Quartiles - Exclusive	Interquartile Range - Inclusive	Interquartile Range - Exclusive
17.5	1.0	70.0	Q1: 18.250000, Q2: 35.500000,Q3:	Q1: 17.750000, Q2: 35.500000,Q3:	34.5	35.5

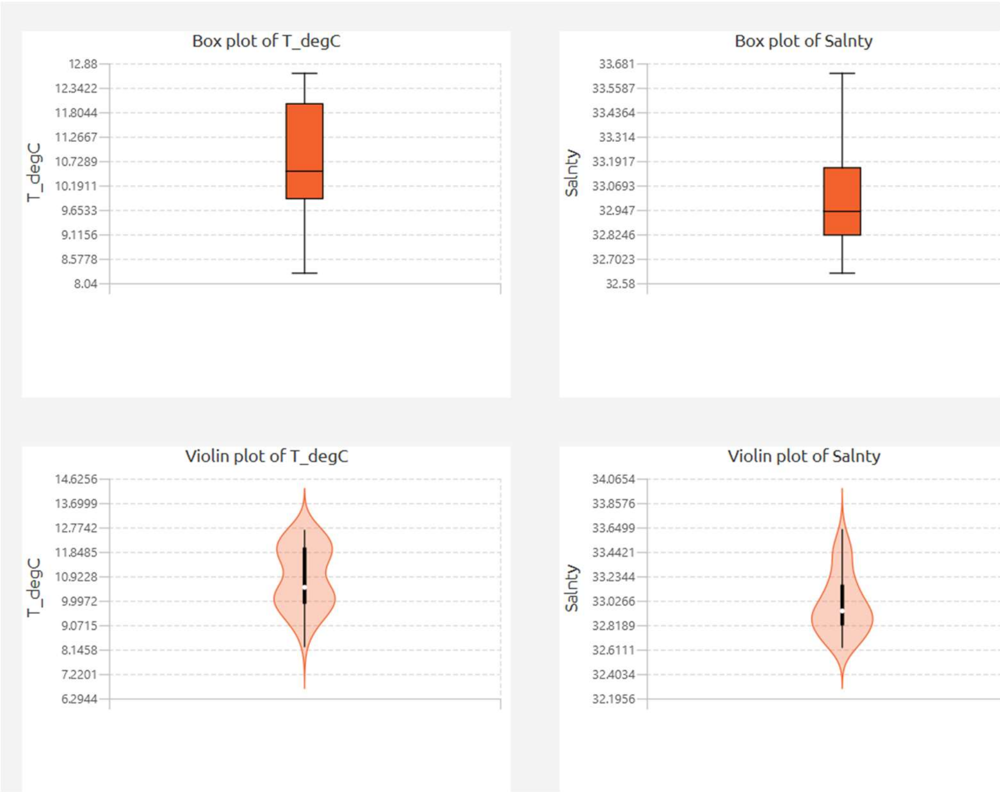
Anderson-Darling: A^2	Anderson-Darling: p-value	Anderson-Darling: Prob	Anderson-Darling: Decision	Kolmogorov-Smirnov: D	Kolmogorov-Smirnov: p-value	Kolmogorov-Smirnov: Prob	Kolmogorov-Smirnov: Decision
0.0626790	0.9299943	0.0700057	Fail to Reject Null Hypothesis	0.7530347	0.0477094	0.9522906	Reject Null Hypothesis

Coefficient Of Variation	Relative Standard Deviation	Root Mean Square	Standard Error Of Mean	Skewness	Kurtosis	Kurtosis Excess	Shapiro-Wilk: W	Shapiro-Wilk: p-value	Shapiro-Wilk: Prob	Shapiro-Wilk Decision
0.5732700	57.3270004	40.847277	2.4324199	0E-7	1.8000000	-1.2000000	0.9550122	0.0133838	0.9866162	Reject Null Hypothesis




Factor: T_degC
Factor: Salnty
Q-Q Plot of Sample Quantiles for factor T_degC
Normal Distribution - 95% CI
Sample Quantiles
Theoretical Quantiles
Data Points
Reference Line
Upper Band
Lower Band
Q-Q Plot of Sample Quantiles for factor Salnty
Normal Distribution - 95% CI
Sample Quantiles
Theoretical Quantiles
Data Points
Reference Line
Upper Band
Lower Band
Histogram of T_degC
Frequency
T_degC
Histogram of Salnty
Frequency
Salnty


Box Plot
Factor: Btl_Cnt
Box plot of Btl_Cnt
Btl_Cnt
Factor: Cst_Cnt
Box plot of Cst_Cnt
Cst_Cnt
Factor: Depthm
Box plot of Depthm
Depthm
Violin Plot
Violin plot of Btl_Cnt
Btl_Cnt
Violin plot of Cst_Cnt
Cst_Cnt
Violin plot of Depthm
Depthm


Box plot of T_degC
T_degC
12.88
12.3422
11.8044
11.2667
10.7289
10.1911
9.6533
9.1156
8.5778
8.04
Box plot of Salnty
Salnty
33.681
33.5587
33.4364
33.314
33.1917
33.0693
32.947
32.8246
32.7023
32.58
Violin plot of T_degC
T_degC
14.6256
13.6999
12.7742
11.8485
10.9228
9.9972
9.0715
8.1458
7.2201
6.2944
Violin plot of Salnty
Salnty
34.0654
33.8576
33.6499
33.4421
33.2344
33.0266
32.8189
32.6111
32.4034
32.1956